

Project Requirement Document

Day 31 | Capstone: Company Knowledge AI Assistant Version: 1.0

1. Project selection and scope

Selected option: **Option 4 - Company Knowledge AI Assistant**. The proposed capstone will answer employee questions about approved company policies, SOPs, FAQs and internal guidance through retrieval-augmented generation (RAG). The solution will build on an existing internship prototype; it will be finalized only after Team Lead review.

2. Problem statement

Company information is often distributed across spreadsheets, documents and informal communication. Employees may need to search multiple locations or contact a colleague to answer recurring policy questions. A generic AI chatbot can respond fluently but may invent company-specific rules. The project aims to provide one conversational entry point that searches an approved knowledge base and returns concise, source-aware answers, or clearly states when information is unavailable.

3. Current process and pain points

Current activity	Pain point / business impact
Employee asks HR or searches documents	Repeated queries and delays in finding the right reference.
Staff manually locates and interprets a policy	Time spent on routine questions; inconsistent wording or outdated copies.
Unknown or ambiguous questions are handled informally	Risk of unsupported answers and unclear escalation ownership.
Updates are made to source material	Knowledge can become stale unless the assistant is refreshed.

4. Proposed solution outline

Trigger and input: An employee opens the n8n chat interface and submits a natural-language question. A chat session identifier supports follow-up context.

Processing: Chat Trigger -> AI Agent -> Supabase vector-search tool -> Gemini-based answer -> chat response. A separate ingestion workflow reads approved source entries, loads/chunks content as needed, creates embeddings and stores them in a Supabase vector table.

AI role: Select the knowledge-search tool; answer only from retrieved material; summarize clearly; include a human-readable source when available; use conversation memory only to interpret follow-up questions.

Business rules: Never invent company policy. If the knowledge base does not explicitly support an answer, return "Information not available in the knowledge base." Ask for clarification when the question is unclear. Do not take sensitive actions automatically.

Storage: Approved knowledge entries and embeddings in Supabase; source content maintained in a controlled source file/sheet. Conversation memory may be used for the demonstration; avoid storing sensitive employee details.

Output and notifications: Return an answer with a source label, clarification question or fallback. For human escalation, provide a clear request-to-contact-HR path; automatic ticket/email escalation is optional and requires an approved destination and workflow.

5. Initial data and scope assumptions

The existing prototype contains **20 sample company-knowledge entries** spanning leave, working hours, attendance, internships, conduct, security, equipment, expenses, meetings and employee support. These are demonstration entries, **not verified Petalnex policies**. Before real-company use, the Team Lead must approve the documents, ownership, access permissions and policy accuracy. Do not present the sample rules as authoritative company policy.

6. Success criteria and acceptance tests

Knowledge ingestion: At least 20 approved or clearly labeled sample entries are ingested and visible in the vector database.

Relevant answers: Run 10 documented queries; answers to covered questions match the source and display a readable source name.

Unknown-answer handling: At least one deliberately unsupported query returns the exact fallback without inventing a policy.

Conversation: A follow-up question in the same session is understood when memory is enabled (bonus).

Escalation boundary: An unresolved question points to human review; no automatic external action occurs without approved configuration.

Reproducibility: Workflow JSON, source files, test results, README and demo are provided, with no secrets embedded.

7. Constraints, dependencies and risks

Dependencies: working n8n, Gemini API quota, Supabase connectivity, embedding-model consistency and access to approved sources. Risks: rate limits (HTTP 429), unavailable services, incorrect retrieval, incomplete source content and stale policies. Mitigations: capture test evidence, label sample data, avoid unsupported answers, document manual re-ingestion and reserve escalation for unresolved cases.